

UNIVERSITY OF MARIBOR
FACULTY OF ELECTRICAL ENGINEERING AND COMPUTER SCIENCE
UM FERİ

1-DOF Series-Elastic Finger Exoskeleton

Actuator Commissioning, Embedded Control, sEMG Intent Detection,
and Virtual SEA Evaluation

Institutional mentor: Prof. dr. Aleš Hacı
Prepared by: Drin Duka
Date: 9 July 2026

Abstract

This project investigated a one-degree-of-freedom series-elastic-actuated (SEA) finger exoskeleton for surface-electromyography (sEMG) intent detection and later sensorimotor analysis. The available Harmonic Drive PMA-8A actuator, Maxon ESCON 50/5 controller, incremental encoder, and STM32 NUCLEO-F446RE were commissioned. Closed-loop low-speed rotation was first verified using the ESCON potentiometer and then commanded by STM32 analog and digital outputs. A laptop bridge was implemented to acquire eight-channel Myo armband EMG and inertial data, calibrate a repeatable pinch-intent template, reject competing gestures and motion, log trials, and communicate with the STM32 over serial.

The physical spring, tendon transmission, finger mechanism, and spring-deflection sensor were unavailable and therefore were not represented as completed hardware. Instead, recorded real EMG classification was replayed through an explicitly labelled virtual SEA model to demonstrate the intended bounded $0\text{--}180^\circ$ motion and the relationship between spring deflection and force. The result is a validated physical actuator-and-embedded-control MVP, a separately validated sEMG intent subsystem, and a reproducible simulation of the missing compliant mechanism. Physical Myo-to-actuator operation, impedance control, and electromechanical-delay measurement remain future integration work.

Contents

1	Objective and Scope	2
2	Laboratory Development Chronology	3
2.1	Evidence handling	4
3	Physical System	5
3.1	Hardware architecture	5
3.2	ESCON configuration and commissioning	6
4	sEMG Intent Detection	8
4.1	Acquisition and features	8
4.2	Calibration and classification	8
4.3	Diagnostic Myo validation	9
5	Virtual SEA Replay	10
5.1	Purpose and assumptions	10
5.2	Replay result	11
6	Results and Limitations	12
7	Conclusion and Future Work	13
A	Reproduction Commands	14

Chapter 1

Objective and Scope

The declared research objective was a 1-DOF dorsal finger exoskeleton driven through a series elastic element. The intended control and sensing chain was

sEMG intent $\rightarrow$ embedded controller $\rightarrow$ motor $\rightarrow$ spring/tendon $\rightarrow$ finger motion.

The proposed series spring had stiffness $k_s \approx 2.0$ N/mm. Its measured deflection would provide an indirect force estimate,

$$F_s = k_s \Delta x_s, \quad (1.1)$$

and, after mechanism calibration, an interaction torque estimate. The intended impedance law was

$$\tau_{cmd} = K_v(\theta_{target} - \theta) + B_v(\omega_{target} - \omega). \quad (1.2)$$

The final available hardware did not include the spring, Bowden/tendon transmission, displacement sensor, or printed finger mechanism. Consequently, this report distinguishes three evidence classes:

- **Measured:** voltages, ESCON monitor values, EMG, IMU, and serial responses;
- **Observed:** physical actuator motion in photographs and videos;
- **Simulated:** finger angle, spring deflection, spring force, and virtual SEA dynamics.

No simulated signal is presented as a physical measurement.

Chapter 2

Laboratory Development Chronology

The final result came from a staged commissioning process rather than a single end-to-end build. This chronology is included because it shows which parts of the original SEA project were physically validated, which parts were deferred, and why the final demonstration scope was reduced.

- **17 June 2026 – hardware familiarisation.** The available PMA actuators, ESCON 50/5 controller, encoder wiring, STM32 board, bench supply, and mechanical stock were inspected. An existing ESCON configuration was opened in ESCON Studio and preserved as a reference. The result was a safe commissioning sequence: identify hardware, prepare cables, verify wiring, power up under supervision, tune, then integrate STM32. No powered motor test was performed.
- **22 June 2026 – cable preparation.** J1 power leads and J2 motor leads were prepared with suitable terminations, insulation, and planned continuity checks. The system remained unpowered.
- **1 July 2026 – STM32 and ESCON setup.** The STM32 toolchain was verified with a NUCLEO-F446RE blink/button project. A fresh ESCON 50/5 was configured for the PMA-8A actuator with conservative current and speed limits, encoder feedback, closed-loop speed control, Analog Input 1 setpoint, and Digital Input 2 enable. The expected VCC undervoltage fault remained because external motor power was not yet connected.
- **2 July 2026 – supervised power-up.** External power was applied under supervision, the undervoltage fault cleared, automatic tuning completed, and Potentiometer 1 was temporarily used as a low-speed setpoint. First controlled actuator motion was achieved near 100 rpm.
- **3 July 2026 – STM32 integration planning.** The STM32 interface was planned as PA4/A2 for DAC speed setpoint, PC0/A5 for digital enable, PC13 for the user button, and PA5 for LED feedback. The initial generated project workflow caused toolchain confusion, so no STM32-to-ESCON motor test was completed that day.
- **4–5 July 2026 – scope recovery.** The plan was reduced to a minimum viable actuator chain: preserve the potentiometer demo, verify STM32 outputs with a multimeter, connect to ESCON only after verification, and keep a fallback demonstration.
- **6 July 2026 – embedded MVP achieved.** A working CubeIDE project named `new-stm-code` was created. PA4/A2 and PC0/A5 were verified independently with a

multimeter, then wired to ESCON Analog Input 1 and Digital Input 2. Pressing the Nucleo button produced about 1.00 V analog command and 3.31 V enable; ESCON demand and actual speed rose near 100 rpm; the PMA actuator rotated repeatably.

- **7–8 July 2026 – Myo and serial diagnostics.** Laptop serial control, Myo armband acquisition, calibration, template matching, gesture rejection, gyro-based motion rejection, live plotting, CSV logging, and virtual SEA replay were added. Myo diagnostic control worked with the motor disabled.
- **9 July 2026 – finalization.** The repository, report, evidence notes, replay plots, and software checks were finalized. The deliverable was frozen as a physical actuator/embedded-control MVP plus diagnostic Myo subsystem and simulated SEA replay.

2.1 Evidence handling

The evidence set contains wiring photographs, multimeter photographs, ESCON Studio screenshots, and short videos of potentiometer-controlled and STM32-button-controlled actuator motion. Large raw phone media were intentionally kept outside Git. The repository includes sanitized report figures, source code, logs, calibration presets, the STM32 firmware tree, and this PDF report. This keeps the repository portable while preserving the evidence chain separately.

Chapter 3

Physical System

3.1 Hardware architecture

The physical test bench comprised a PMA-8A-50-01-E500ML actuator, ESCON 50/5, incremental encoder, NUCLEO-F446RE, and an external DC supply. The ESCON handled encoder-based closed-loop velocity control. The STM32 supplied only the speed setpoint and enable signal.

Table 3.1: Verified STM32-to-ESCON interface.

STM32 signal	ESCON connection	Function
PA4 / A2	J6 pin 1, Analog Input 1+	Analog speed setpoint
GND	J6 pin 2, Analog Input 1-	Analog reference
PC0 / A5	J5 pin 2, Digital Input 2	High-active enable
GND	J5 pin 5, Signal GND	Digital reference

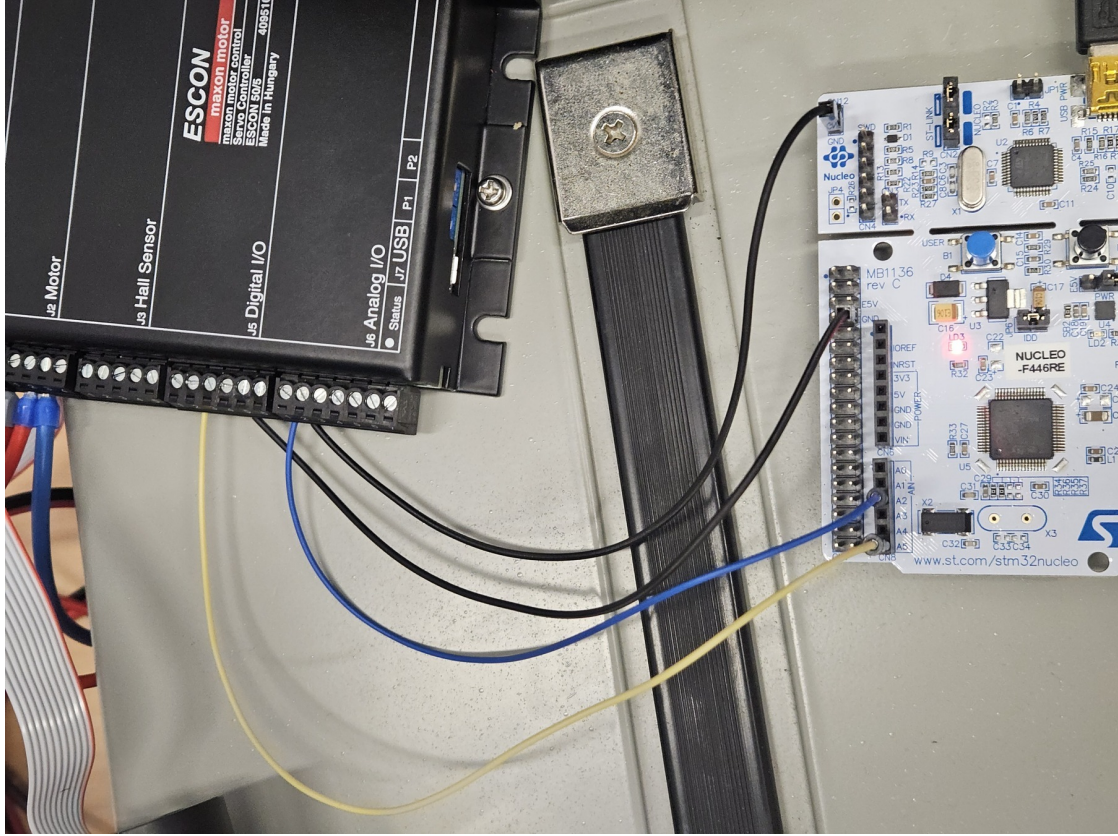


Figure 3.1: Observed STM32-to-ESCON signal wiring. The photograph documents the physical command interface, not a completed finger mechanism.

3.2 ESCON configuration and commissioning

The ESCON was configured as a closed-loop speed controller with a 500-pulse digital incremental encoder. Initial limits were deliberately conservative.

Table 3.2: Commissioning configuration recovered from the final ESCON file.

Parameter	Value
Controller	Maxon ESCON 50/5
Control mode	Closed-loop speed control
Setpoint source	Analog Input 1
Setpoint scaling	0 V $\rightarrow$ 0 rpm; 1 V $\rightarrow$ 100 rpm
Enable	Digital Input 2, high-active
Encoder	Digital incremental, 500 pulses/revolution
Nominal current	0.60 A
Maximum output current	1.00 A
Acceleration/deceleration	100 rpm/s
Maximum permissible speed	6000 rpm

Supervised automatic tuning completed successfully. Stable rotation was first observed using Potentiometer 1. The STM32 interface was then verified independently with a multimeter before connection to the ESCON. The measured active outputs were approximately 1.00 V on PA4 and

3.31 V on PC0; inactive output was approximately 0.00 V.



Figure 3.2: Measured STM32 command voltages before connection to the motor controller: inactive DAC (left), active DAC (centre), and active digital enable (right).

Pressing the Nucleo user button asserted enable and the 1 V command; ESCON Studio showed demand and actual speeds near 100 rpm, and the actuator rotated stably. Releasing the button returned the DAC to 0 V, removed enable, and stopped the actuator. Repeated start/stop operation was recorded on video. This is the completed physical MVP.

Chapter 4

sEMG Intent Detection

4.1 Acquisition and features

The deprecated Myo armband was connected to a laptop over BLE. The laptop subscribed to eight EMG channels and IMU data. No Bluetooth discovery was required during normal use because the known device address was supplied directly. For each 200 ms sliding window with a 50 ms step, the mean absolute value (MAV) was calculated:

$$f_i = \frac{1}{N} \sum_{n=1}^N |e_i[n]|, \quad i = 1, \dots, 8. \quad (4.1)$$

Total activity and the normalized spatial pattern were

$$A = \sum_i f_i, \quad p_i = \frac{f_i}{A + \epsilon}. \quad (4.2)$$

4.2 Calibration and classification

Calibration recorded rest in three forearm orientations, several candidate intents, static rejection gestures, and wrist/forearm motion. Candidate intents were ranked by within-class cosine similarity minus their strongest rejection similarity. The selected pinch attempt was retained only when it produced sufficient activity above rest and a repeatable spatial pattern.

Live classification required activity above the orientation-specific rest threshold, adequate target similarity, a positive margin over the strongest rejection class, and low gyroscope magnitude. Hysteresis and hold times suppressed rapid switching. The detector states were REST, TARGET, GESTURE_REJECT, and MOTION_REJECT. TARGET generated serial command 1; all other states generated 0. A three-second STM32 command timeout and repeated STOP commands provided communication fail-safe behaviour.

Calibration presets can be reused, numerically adjusted, or partially retaken. Every run logs timestamped raw EMG, MAV features, similarity scores, state, command, IMU magnitude, and virtual-model outputs to CSV.

4.3 Diagnostic Myo validation

The final diagnostic session on 8 July 2026 ran for about 167 s with the motor command disabled. The log contained rest periods, repeated pinch attempts, open-hand rejection, full-fist rejection, wrist flexion/extension, and forearm rotation. Of 1611 logged windows, 384 were classified as REST, 91 as TARGET, 567 as GESTURE_REJECT, and 569 as MOTION_REJECT. This is not presented as a robust clinical classifier; it is evidence that the software can accept the calibrated pinch intent while suppressing many competing gestures and high-motion intervals.

The strongest limitation observed in the log was that pinch attempts can be interrupted by wrist/forearm motion rejection, and open-hand or full-fist templates can sometimes remain close to the target template. Therefore, the correct interpretation is “diagnostic intent interface proven,” not “finished proportional hand controller.” Motor-enabled Myo testing was deliberately postponed until the mechanism and safety limits are better defined.

Chapter 5

Virtual SEA Replay

5.1 Purpose and assumptions

The virtual model demonstrates the missing compliant mechanism without claiming physical validation. It replays a real recorded Myo session. Pinch intent commands a target of 0° and non-target/relaxed state commands 180° . Motion is bounded to this interval and rate-limited to $90^\circ/\text{s}$.

The virtual motor angle follows the command subject to the rate limit. Finger angle follows motor angle with a first-order time constant $\tau = 0.18$ s. With an assumed 8 mm tendon moment arm,

$$x_m = r(\theta_{max} - \theta_m), \quad (5.1)$$

$$x_f = r(\theta_{max} - \theta_f), \quad (5.2)$$

$$\Delta x_s = |x_m - x_f|, \quad (5.3)$$

$$F_s = \min(k_s \Delta x_s, F_{max}), \quad (5.4)$$

where angular quantities are converted to radians, $k_s = 2.0$ N/mm, and $F_{max} = 5$ N. These values are adjustable simulation parameters, not identified properties of an assembled mechanism.

5.2 Replay result

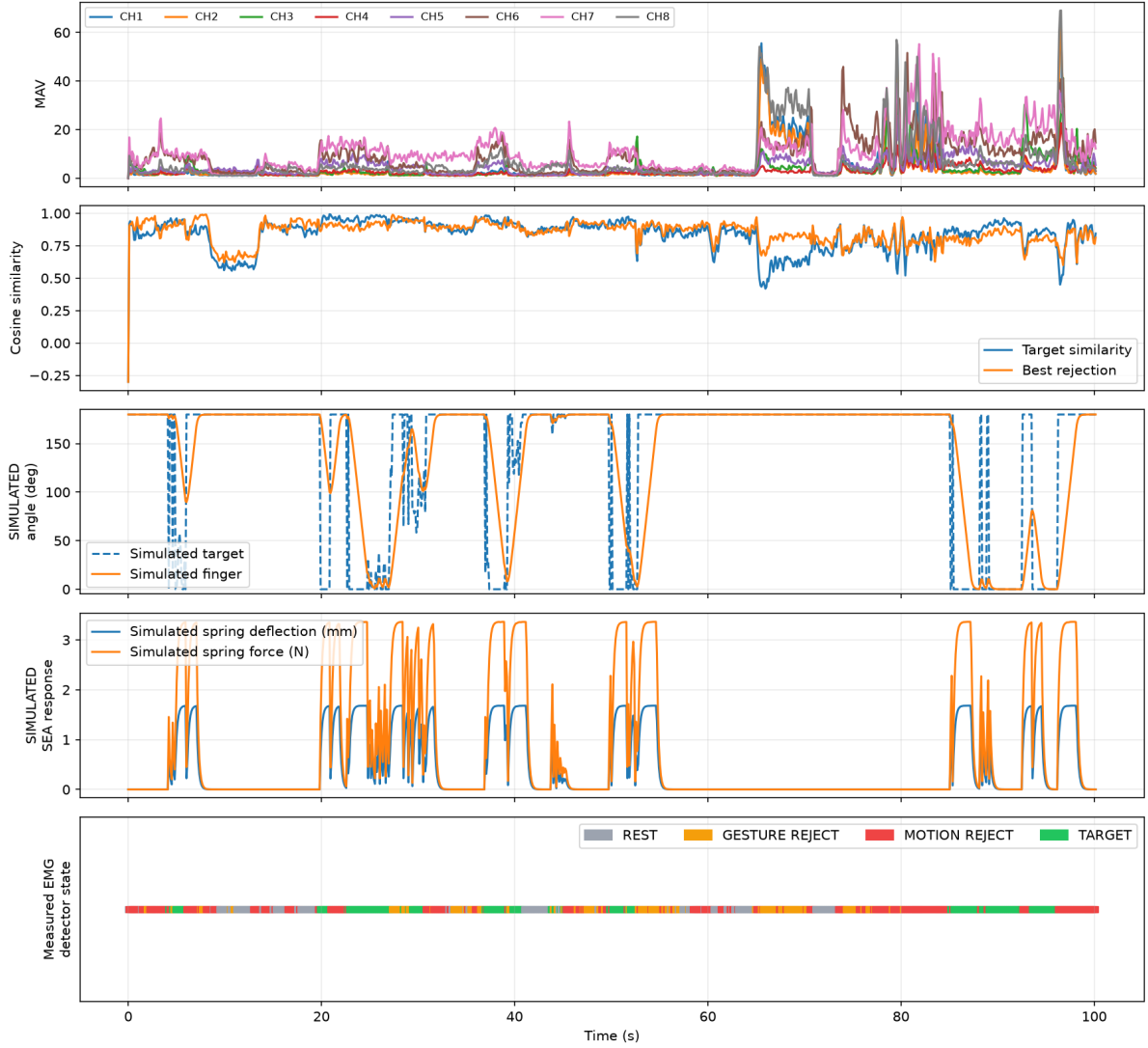


Figure 5.1: Replay of recorded real Myo data. MAV, cosine similarities, and detector states are measured/processed data. Finger angle, spring deflection, and force are explicitly simulated.

The replay shows that accepted pinch intervals produce bounded virtual flexion while rejected gestures and high-motion intervals do not sustain the flexion command. Transient motor–finger displacement produces virtual spring deflection and force. This verifies software data flow and intended qualitative behaviour; it does not validate physical impedance, force accuracy, or human interaction safety.

Chapter 6

Results and Limitations

Table 6.1: Status against the declared study.

Item	Status	Evidence
Actuator and ESCON commissioning	Completed	Tuning, potentiometer motion, monitor screenshots, video
STM32 analog command and enable	Completed	Multimeter measurements, wiring photographs
STM32-controlled motor motion	Completed	Repeated button start/stop video
Myo acquisition and intent detection	Completed	Calibration, terminal output, CSV logs, plots
Laptop-STM32 serial protocol	Completed	ACK/STATUS transcript and fail-safe firmware
Physical Myo-to-motor experiment	Not performed	Kept disabled pending controlled safety test
Spring, tendon, and finger mechanism	Not implemented	Components unavailable
Spring-deflection force sensing	Simulated only	Virtual replay
1 kHz physical impedance control	Not implemented	Requires complete SEA sensing and mechanics
EMD/reflex experiment	Not performed	Requires measured spring-force onset and approved protocol

The ESCON is a velocity controller, not an absolute-position controller. Its internal encoder closes the velocity loop but does not provide the STM32 with verified finger angle. Therefore the present physical system cannot safely claim 0–180° finger positioning. A physical SEA requires the compliant element, a calibrated deflection sensor, mechanism geometry, travel limits, and preferably independent output-position sensing.

Chapter 7

Conclusion and Future Work

The project established a reproducible actuator-control baseline and extended it with calibrated Myo intent detection. The physical MVP proves that the STM32 can safely command the ESCON and PMA actuator. The software MVP proves that a repeatable pinch intent can be separated from rest, competing gestures, and high-motion activity for this user. The replay connects recorded EMG intent to an explicitly virtual compliant finger model while preserving the boundary between evidence and simulation.

The next engineering step is mechanical rather than algorithmic: fabricate the 1-DOF finger and tendon path, select and calibrate the spring-deflection sensor, add output travel sensing or hard limits, and only then implement and tune the physical impedance loop. EMD measurement should follow validation of synchronized EMG and measured spring-force onset.

Appendix A

Reproduction Commands

Run the Myo system without a motor command:

```
.venv/bin/python run_myo_sea_demo.py \  
  --myo-address D4:71:C5:FF:86:F3 \  
  --motor disable --mode diagnostic --live-plot
```

Generate the measured/simulated replay figure:

```
MPLBACKEND=Agg .venv/bin/python plot_log.py logs/SESSION.csv \  
  --output report/figures/myo-sea-replay.png
```

Run the software checks:

```
.venv/bin/python -m unittest -v  
.venv/bin/python sea_model.py
```


Bibliography

- [1] G. A. Pratt and M. M. Williamson, “Series Elastic Actuators,” *IEEE/RSJ IROS*, 1995.
- [2] N. Hogan, “Impedance Control: An Approach to Manipulation,” *Journal of Dynamic Systems, Measurement, and Control*, 1985.
- [3] Maxon Motor AG, *ESCON 50/5 Hardware Reference*, document 409510.